Diseases of the Nail Unit

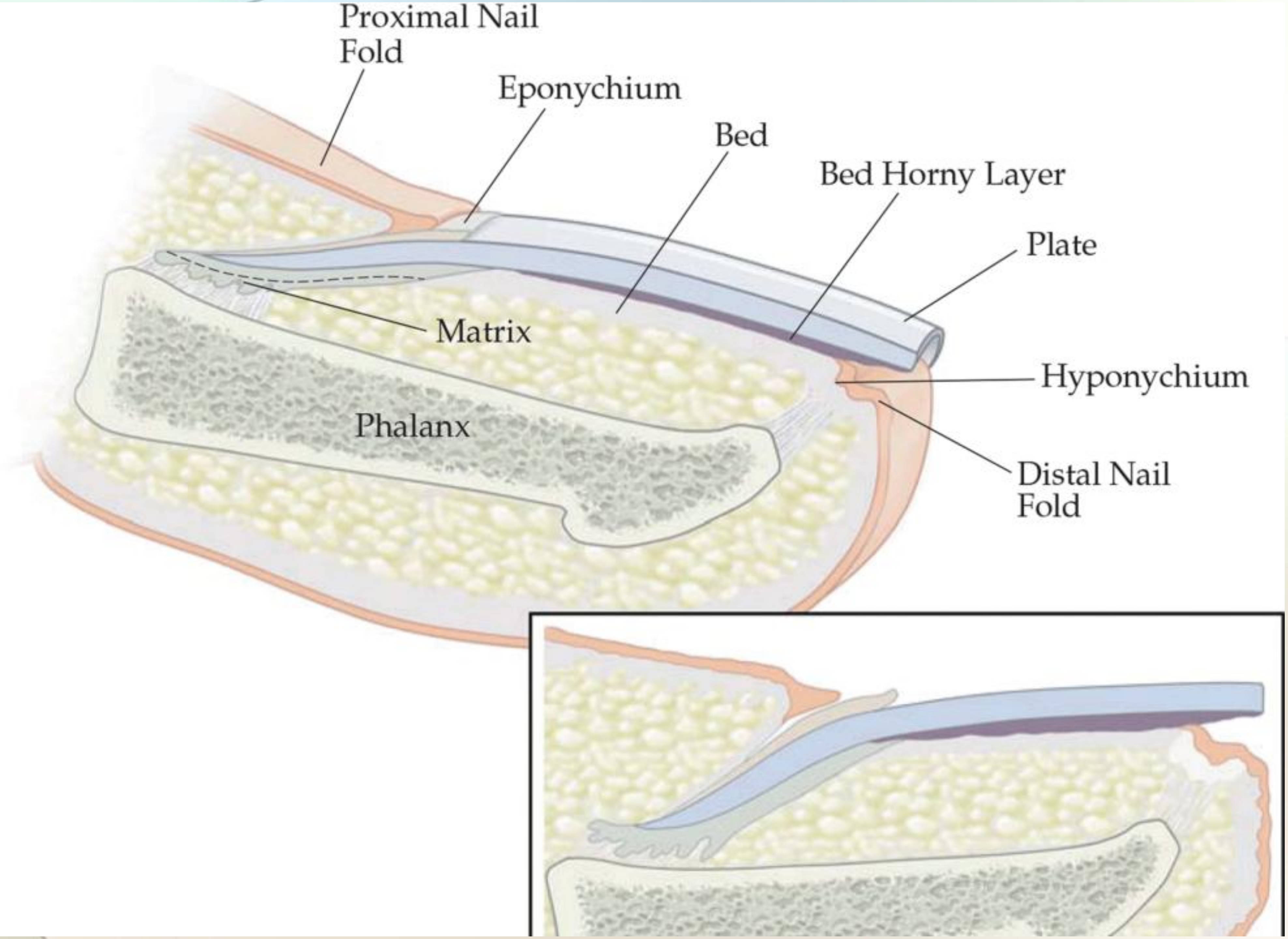
Objectives

- Review function and structure of nail components
- Review nail findings associated with systemic and dermatologic conditions
- Become familiar with diagnosis and treatment of infections of the nail.

Introduction

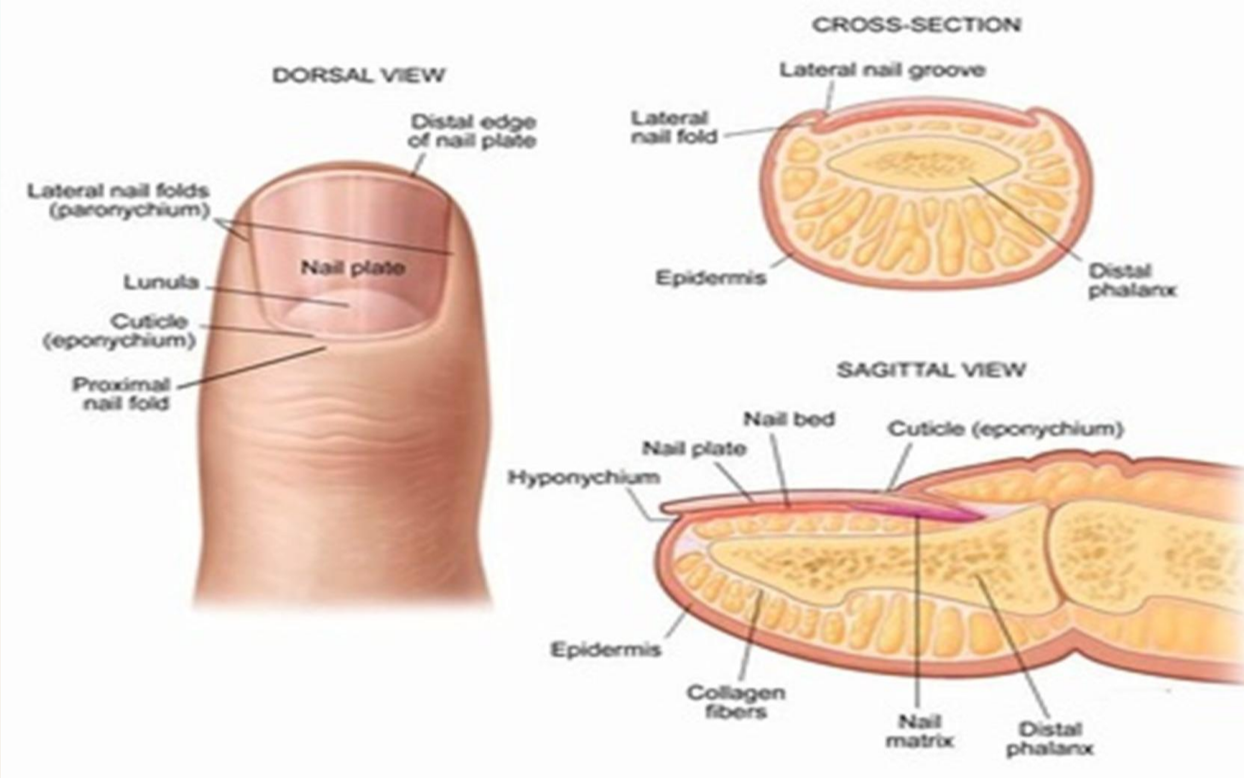
- Due to their visibility, functional importance, marked acuity of fingertip sensation, and aesthetic significance, especially with fingernails, disorders affecting the nails can have a profound adverse impact on the quality of life (QOL) of affected individuals.
  - When severe, and can interfere with interpersonal interactions and work ability in some cases.

Nail Structure, Function and General Pathophysiologic Concepts


Proximal Nail
Fold
Eponychium
Bed
Bed Horny Layer
Plate
Matrix
Hyponychium
Phalanx
Distal Nail
Fold

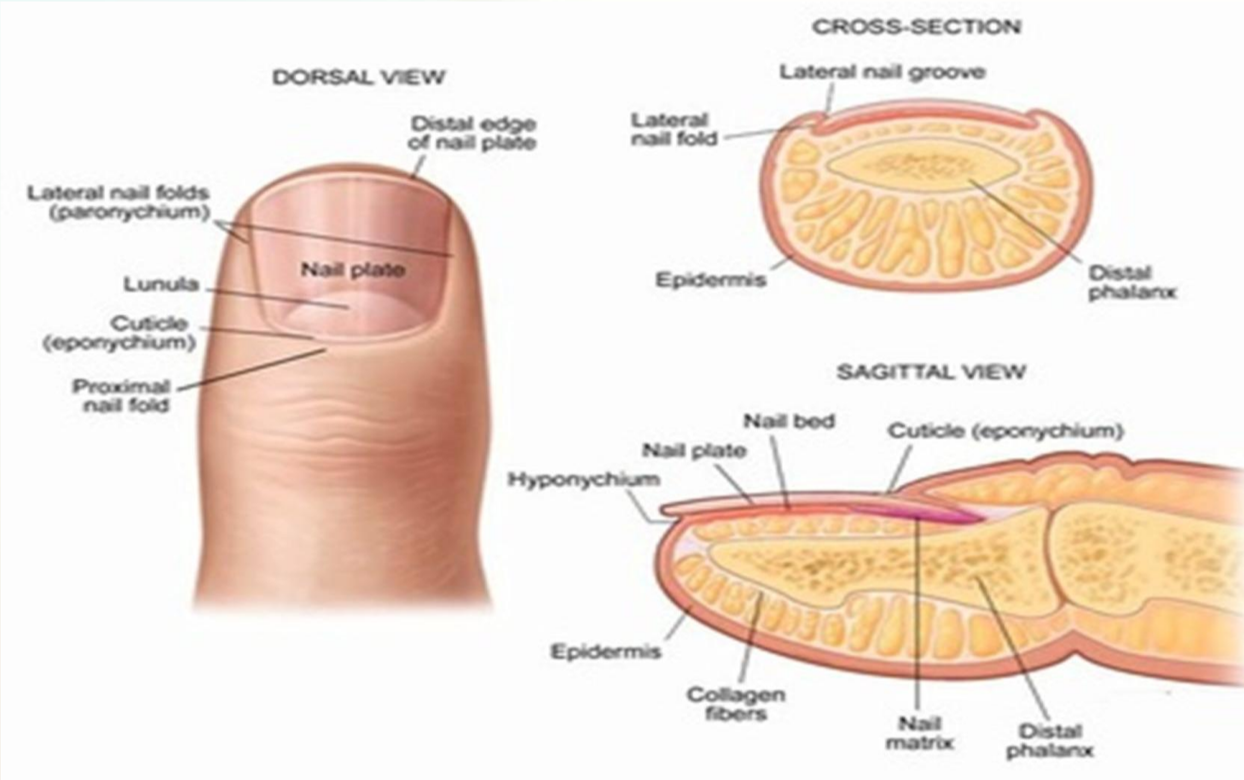
Anatomy:

- Matrix - the dynamic, germinative portion of the nail unit that produces the nail plate.
- The lunula is the visible portion of the nail matrix, appearing under the proximal nail plate as a gray-white half-moon projecting just distal to the proximal nail-fold cuticle.
  - Discoloration suggests a pathologic process.



Anatomy:

- Nails are usually devoid of pigmentation because of the relatively sparse number of melanocytes present in matrix epithelium
- The presence of a solitary longitudinal pigmented band of the nail plate, especially if new, irregular in shape, and not uniform in color, is suggestive of a melanocytic process, which could be atypical or malignant.



...although can be normal

Multiple pigmented longitudinal bands in Fitzpatrick skin type V and VI

If pigment involves the skin of the proximal or lateral nail fold (aka periungual pigmentation, aka Hutchinson sign) --> think Melanoma

Nail Matrix

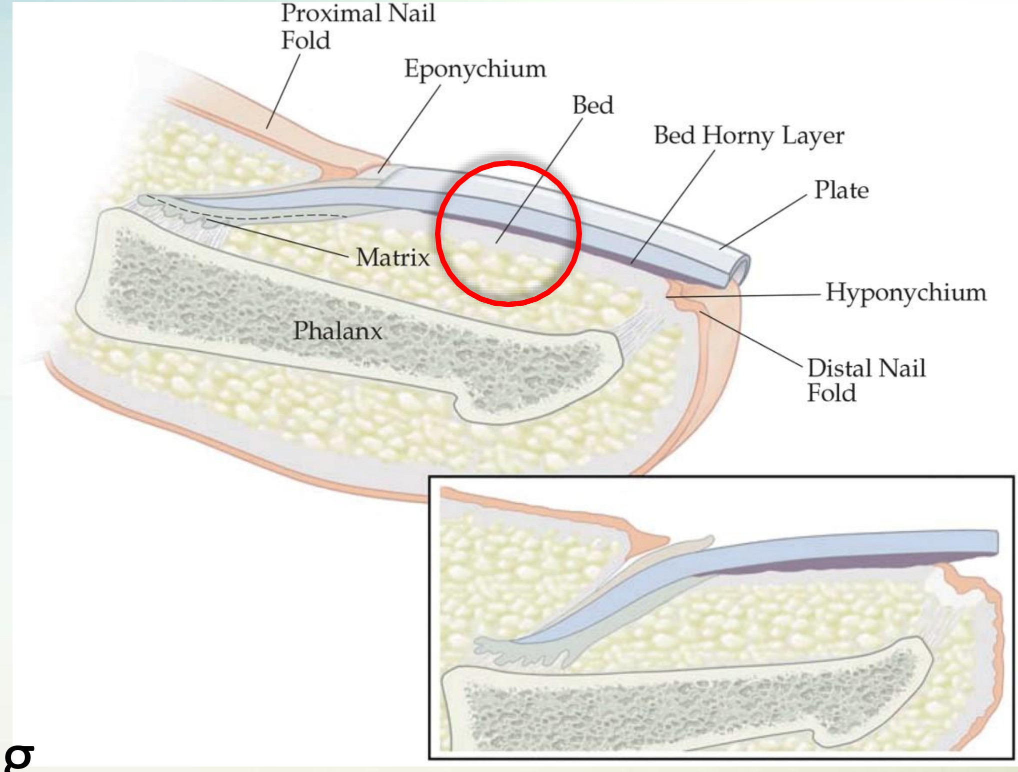
- Diseases of the distal nail matrix result in abnormalities of the undersurface of the nail plate, changes that are visible at the free edge of the nail, or both.
- Disorders of the proximal matrix produce surface abnormalities of the nail plate.
  - Characteristic example is nail-plate pitting secondary to psoriasis

Nail Matrix

- Permanent damage to the matrix may result in permanent nail-plate dystrophy.
- Some disorders cause inflammation of the nail matrix initially, which progresses to produce scarring with permanent destruction of the nail unit

Nail Bed

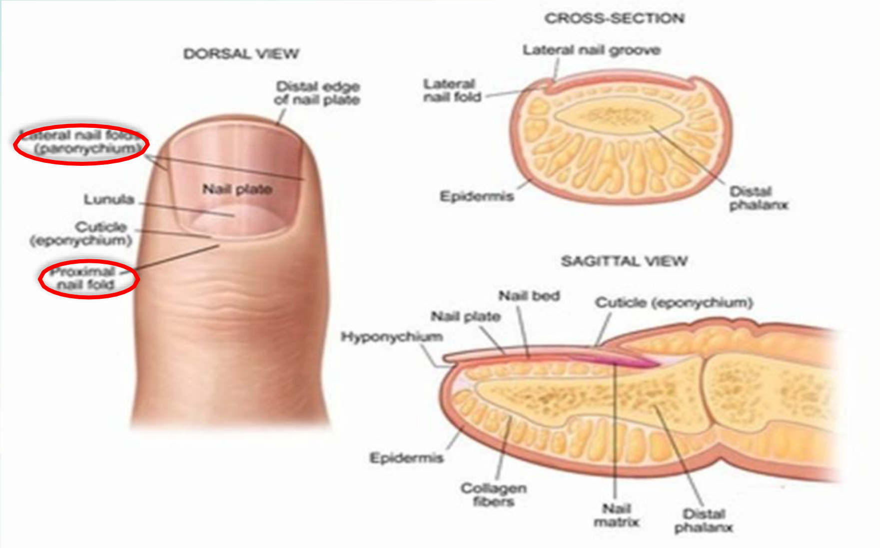
- layer of epithelium lying between the lunula and the hyponychium (the distal epithelium at the free edge of the nail).
- The surface epithelium of the nail bed is longitudinally ridged, with small, superficially oriented vessels, interdigitating with an array of ridges on the undersurface of the nail plate.
  - This anatomic feature explains the longitudinal linearity of splinter hemorrhages, which are foci of extravasation wedged between the bed and the plate.
    - As outgrowth of the nail plate occurs, splinter hemorrhages progress distally.



Nail Plate

- Growth rate of an adult fingernail is about 3 mm/month individuals
  - Toenail growth occurs at one third to one half the rate
- Nail-plate growth is faster in children, peaking between 10 and 14 years of age
- Growth increases during pregnancy
- Growth decreases during lactation, after use of chemotherapeutic agents, and in conditions characterized by limb paralysis, persistently diminished circulation, or malnutrition.

Nail Folds



- The nail folds are the cutaneous soft tissue that house the nail unit, invaginating proximally and laterally to encompass the emerging nail plate.
- The term *paronychia* describes inflammation of the nail folds.
  - may be acute or chronic and may occur secondary to a variety of conditions, including contact dermatitis, psoriasis, bacterial infection, and fungal infection
- The cuticle (eponychium) is a thin, keratinized membrane of modified stratum corneum that extends from the distal portion of the nail fold, reflecting onto the nail-plate surface.
  - Intact cuticle serves as a seal that protects the space between the nail folds and the nail plate from exposure to external irritants, allergens, and pathogens.
- Inflammatory paronychia is most often secondary to irritant contact dermatitis from exposures to several contactants that have gained access under the exposed nail folds, which are normally protected by an intact cuticle.

Table 2 Selected Dermatologic Disorders Affecting the Nail Unit		
Disease State	*Disease Features*	*Nail Findings*
Inflammatory diseases Psoriasis	Nail findings in 10–50% of patients; 39% of children with psoriasis with nail changes (usually pitting); nail disease present in 50–85% of patients with psoriatic arthritis	Proximal matrix involvement: pitting, transverse grooving, deeply ridged plate surface (onychorrhexis) Distal matrix involvement: plate thinning, lunula erythema Nail bed: subungual hyperkeratosis, oil drop sign, splinter hemorrhages Nail folds: cutaneous lesions of psoriasis Phalangeal/joint involvement: psoriatic arthritis
Lichen planus	Nail changes occur in up to 10% of patients with lichen planus; may occur in childhood or adulthood; nail involvement may be present with or without skin or mucosal disease; potentially reversible in early inflammatory stage; irreversible in cicatricial (later stage) of disease; may present as ridged, rough-surfaced, lusterless plates (trachyonychia) or 20-nail dystrophy in children	Matrix involvement: combination of nail-plate ridging, splitting, and progressive uniform thinning; distal-edge splitting, fragility, crumbling, brittleness, nail-plate shedding (onychomadesis) Focal matrix scarring: pterygium formation (scarring bridge between proximal nail fold and subungual epidermis with focal loss of nail plate) Nail-bed involvement: subungual hyperkeratosis, onycholysis Diffuse matrix/nail-bed disease: total nail-plate loss, atrophy, scarring
Alopecia areata	Nail changes in 10% of patients with alopecia areata; nail changes in over 40% of children with alopecia areata; fingernail involvement most common; may present in children as 20-nail dystrophy	Matrix involvement: orderly nail pitting arranged in a cross-hatched pattern (glen plaid sign); roughened nail-plate surface (trachyonychia); fragility; splitting; longitudinal ridging; spotted or red lunula (erythema); nail-plate shedding (onychomadesis)
Nail tumors Glomus tumor	75% occur on the hand, usually subungual (nail bed); a benign vascular hamartoma	Visible through plate as a light-red, reddish-blue spot; rarely exceeds 1 cm in size; characteristic symptom of intense or pulsatile pain; pain is spontaneous or provoked by slight trauma or pressure
Digital myxoid (mucus) cyst	A form of focal mucinosis; not a true cyst (no epithelial lining); contains clear, viscous, jellylike fluid; usually seen in adults	Soft, domed, translucent, pink or skin-colored, shiny neoplasm of proximal nail fold or overlying distal interphalangeal joint; those over fold may compress matrix, producing flattening of plate; those over joint may connect to underlying joint space
Subungual exostoses	Outgrowths of calcified cartilage or normal bone; most seen on great toe; most frequent in adolescents and young adults; benign lesions	Emerge from the dorsal digit at distal phalanx; may erode through plate or project from under distal or lateral edge of plate; often painful; may become eroded
Periungual angiofibromas	Arise out of nail fold; often multiple; seen in 50% of cases of tuberous sclerosis (Bourneville-Pringle disease); usually arise in early teenage years; benign neoplasm	Small, round, flesh-colored or pink, firm papules with shiny, smooth surface arising from nail-fold region; may partially cover nail plate; usually asymptomatic

Nail Findings Associated with Selected Disease States

Infectious:

Onychomycosis (fungal infection)

- Tinea unguium - 50% of all nail infections; nail can turn white, yellow, black or green, thicken, become chalky, and crumble (caused by *Trichophyton rubrum* or *T. mentagrophytes*)
- These fungi cause ringworm of the skin
- *Candida* onychomycosis is far less common than dermatophyte onychomycosis.
  - When present, it is more common in fingernails than toenails and also exhibits nail-plate invasion.

Onychomycosis (fungal infection)

- Treatment: First line --> **Oral** Terbinafine
- Topical antifungals are generally poorly effective due to poor penetration of the nail plate (although may use if there's contraindications to using oral treatment)

Paronychia (Infection involving the nail folds)

- Staph or Strep infection often associated with nail biting, manicuring, or frequent water immersion
- Treatment requires drainage of a focal abscess, if present, and oral antibiotic therapy.

Onychia (inflammation and pus around nail folds)

Associated condition:

- Microscopic pathogens that enter through small wounds
- Often associated with ingrown nails

Inflammatory:

Nail-Plate Pitting

- Psoriasis may produce a random array of shallow or deep pitted indentations, usually affecting one or more fingernail
- Nail pitting may also be seen in association with alopecia areata, punctate keratoderma, and idiopathic trachyonychia
- Occurs in one third of children with alopecia areata often presenting as orderly rows of shallow, pitted depressions.
    - Currently, there is no available treatment for this type of nail pitting.

Onychorrhexis (brittle nails)

Associated Conditions:

- Atopic Chemical exposure (strong soaps, nail polish remover, formaldehyde, etc)
- Hypothyroidism
- Anemia
- Malnutrition
- Oral retinoid therapy (treatment for keratinization disorders)
- Darier's Disease (genetic; associated with skin rash)

Onychogrphosis - thickened nails

- Fingernails demonstrate a tendency to become thinner and more fragile over time
- Toenails usually become thicker and harder.
- Onychogryphosis is a marked thickening, usually of the large toenail, resulting in a compacted mass of heaped-up dystrophic nail plate.
  - Contributing factors appear to be advanced age, poor nail care, chronic trauma, decreased peripheral circulation, poor fitting shoeware, neuropathy.

Onychoschiza (Nail Splitting)

Associated Conditions:

- Psoriasis
- Folic Acid deficiency
- Vitamin C deficiency

Trachyonychia (Sand-paper nails)

- Sandpaper/dull appearance
- Seen in:
  - Psoriasis
  - Chemical exposures
  - Lichen planus
  - Autoimmune diseases
  - Tight fitting shoes

Cuticle Invasion - trachyonychia

Cuticle Invasion - Pterygium

Associated Condition:

- Lichen planus
- Trauma
- Steven Johnson Syndrome

Other:

Distal Onycholysis (detachment from the the nail bed)

Associated Conditions:

- Fungal infection
- Trauma / excess manicuring
- Hyperthyroidism
- Peripheral ischemia
- Photosensitive drug reactions
- Atopic dermatitis
- Amyloidosis
- Connective tissue disorders
- Psoriasis
- Bronchiectasis
- Syphilis
- Hyper and hypothyroidism
- Reynaud
- Hepatocellular dysfunction

Onycholysis

- When moisture accumulates under an onycholytic nail plate, bacterial proliferation may occur.
- This can cause a green discoloration of the nail plate as a result of a pigment produced by certain organisms (e.g., *Pseudomonas aeruginosa*)

Onycholysis

- Treatment - with great subungual discomfort
  - requires avoidance of precipitating factors
  - débridement of the separated nail plate
  - twice-daily topical application of diluted acetic acid solution (consisting of equal parts of white vinegar and water), gentamicin, or the combination of polymyxin B and bacitracin.

Trauma:

Onchogryphosis (Ram's Horn nails)

Associated Conditions:

- Peripheral Vascular Disease
- Trauma
- Failure to cut nails

*Often seen in the elderly

Subungual Hematoma

Associated Condition:

- Blood pools under the nail(hematoma) due to acute injury or repetitive
- Painful

*Note: treatment involves drilling a hole in and releasing the blood

Onychoptosis (nail shedding)

Associated conditions:

- Trauma
- Syphilis
- Hand Foot and Mouth disease
- Fever
- Drug reactions

Deficiencies:

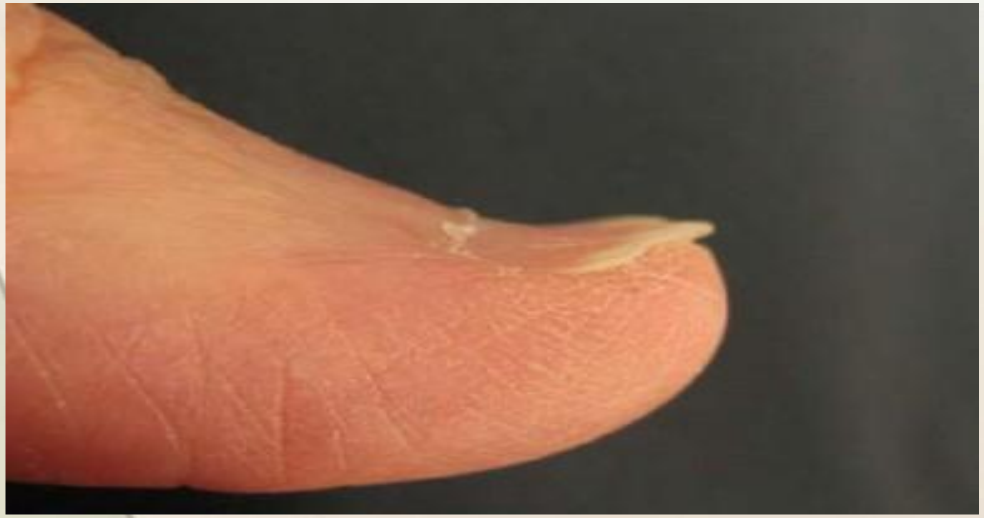
Koilonychia (spooning)

Associated Condition:

- Iron deficiency anemia
- Protein deficiency especially in sulfur-containing amino acids (cysteine or methionine)

Koilonychia

- Water drop test: imagine placing a drop of water on the nail. If it would not fall off it is spooned



Leukonychia (white spots)

Associated Conditions:

- Vigorous and repeated manicuring
- Zn deficiency
- Calcium deficiency
- Sickle cell anemia

Leukonychia

- White discoloration of the nail plate or subungual tissue,
- has multiple presentations.
  - Small 1 to 3 mm white spots (punctate leukonychia) or irregular transverse streaks (leukonychia variegata) of the nail plate are the most common varieties.
- generally secondary to repeated microtrauma to the matrix, growing out distally with outgrowth of the nail plate.
- Mees lines - transverse 1 to 2 mm white bands, usually demonstrated at the same site in multiple nails and occur in association with arsenic intoxication, Hodgkin disease, sickle cell anemia, renal failure, and cardiac insufficiency.
- Leukonychia is also associated with systemic infection and chemotherapy.

Leukonychia

- Half-and-half nails (Lindsay nails) present as a diffuse, dull whitening of the proximal nail bed that obscures the lunula and as a distal region of pink or reddish- brown discoloration that occupies from 20 to 60% of the nail length.
- The most commonly reported association with half-and- half nails is chronic renal failure.
- When the distal brown band of discoloration constitutes less than 20% of the total nail length, the anomaly is known as Terry nails, which occur in association with chronic congestive heart failure, hepatic cirrhosis, type 2 diabetes mellitus, and advanced age.
- In both half-and-half nails and Terry nails, the proximal portion of the nail bed may be light pink, exhibiting a more normal appearance, rather than white.

Leukonychia

- Muehrcke nails present as paired, white, narrow transverse bands of the nail bed separated by normal-appearing thin pink bands
- associated with chronic hypoalbuminemia and edema of nail plate
- Resolution of this nail finding correlates with normalization of serum albumin levels.

Vertical Ridges

Associated Conditions:

- Hypochlorhydria (associated with aging)
- Rheumatoid arthritis (more severe ridging)

Nail beading

- Endocrine Conditions: DM, thyroid disease, Addison’s

Central nail ridge

- Causes:
  - Iron deficiency, folic acid deficiency, protein deficiency

Central nail canal/heller's fir tree deformity

- Cuticle usually normal
- Associations:
  - Severe arterial disease
  - Severe malnutrition
  - Repetitive trauma

Genetics:

Ectodermal dysplasia / Anonychia congenital (absence of nails)

- Genetic disorder affecting the development of teeth, hair, nails, and sweat glands)

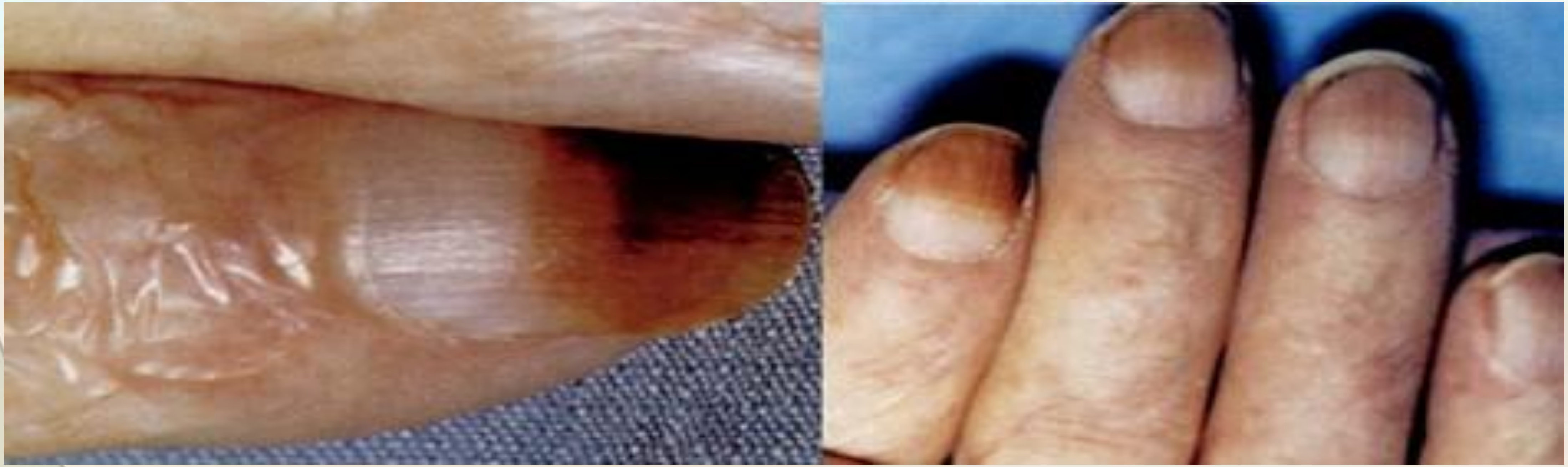
Mental / Emotional:

Bitten nails

- Onychotillomania – compulsive picking or tearing off of nails
- Onychophagia – nails are bitten and chewed off due to anxiety
- Dermatillomania – the skin around the nail is bitten or scratched

Quitter's nail

- Distal nicotine stains with demarcation and no staining proximally seen when a patient quits smoking or changes to a lower tar tobacco.



More serious conditions:

Brown-gray Nails (Melonychia)

Associated Conditions:

- Cardiovascular disease
- Diabetes mellitus
- Vit B12 def
- lichen planus
- Syphilis
- Atopic dermatitis (hair dye, varnish, formaldehye, etc)
- Inflammatory disorders
- Breast cancer
- Malignant melanoma
- Addison's
- Nicotine use (smoking)

Green-black nails (chromonychia)

Associated Condition:

- *Candida* infection
- *Pseudomonas* infection (m/c is *P. aeruginosa*)

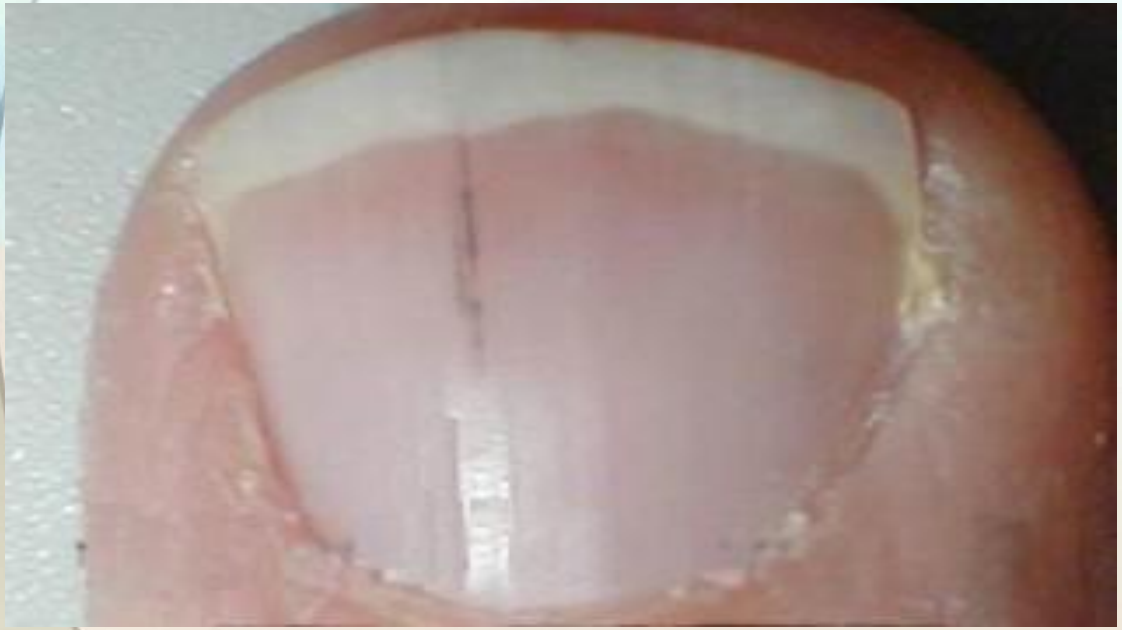
*note: black is most common associated with pseudomonas

Dark longitudinal lines on nails

Associated Conditions:

- Repetitive trauma (running in small shoes)
- Systemic and inflammatory disorders
- Trauma
- Fungal infections
- Drug reaction
- Subungual melanoma
- Benign melanocytic hyperplasias

Splinter hemorrhage

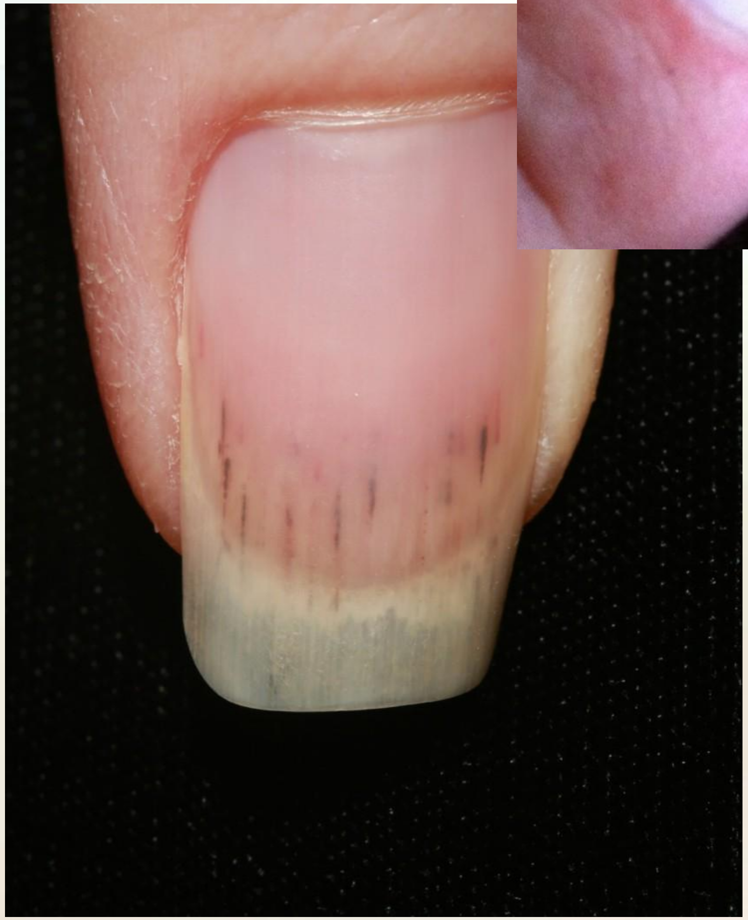


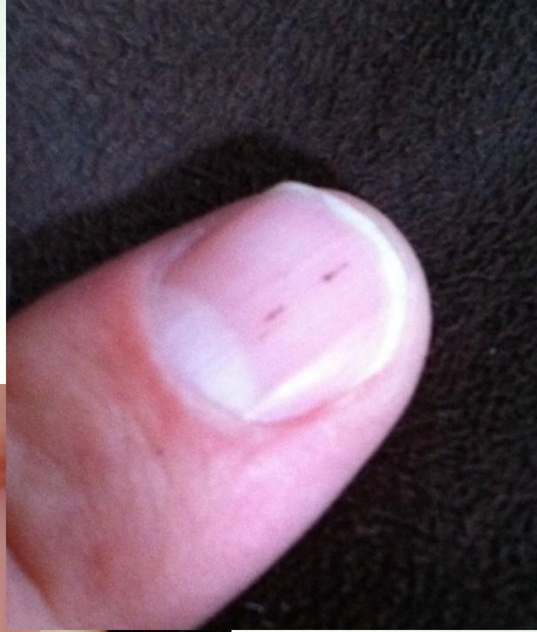
Associated Conditions:

- Bacterial Endocarditis (if left untreated can lead to heart failure)

Splinter Hemorrhages

- The simultaneous appearance of splinter hemorrhages in several nails should raise suspicion of a possible underlying systemic disorder, especially in female patients





Red Nails

Associated Conditions:

- Polycythemia (dark red)
- Lupus
- Carbon Monoxide (cherry red)
- Angioma
- Malnutirion

Longitudinal Erythronychia (red bands)

Associated conditions:

- Dariers Disease
- Lichen Planus

- Differentiate from melanoma

Yellow Nails

Associated Conditions:

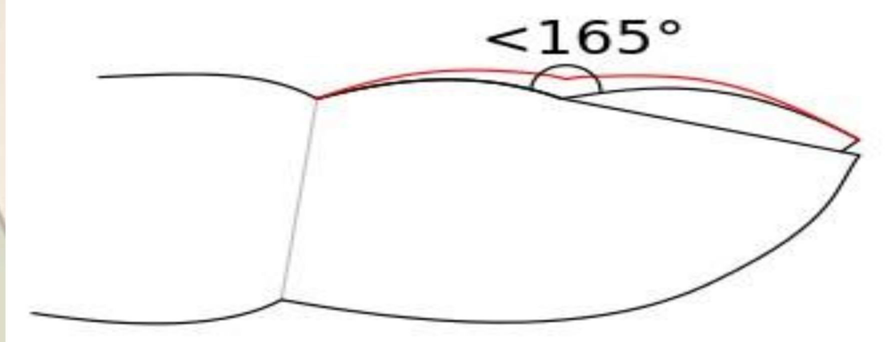
- Yellow nail syndrome (associated with swelling in other parts of the body)
- Amyloidosis
- Jaundice
- Diabetes mellitus
- Median/ulnar nerve injury
- Bronchiectasis
- Thermal injury

Beau's Lines
(horizontal ridges in the nail plate)

Associated Conditions:

- Systemic conditions like Raynauds (especially after cold exposure)
- Pemphigus
- Trauma
- Uncontrolled diabetes
- Infectious: scarlet fever, measles, mumps, pneumonia

Clubbing



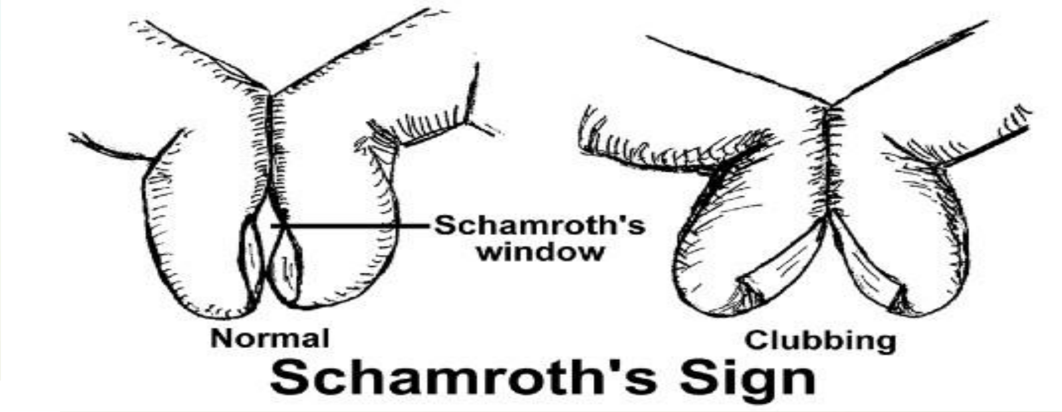
Associated Conditions:

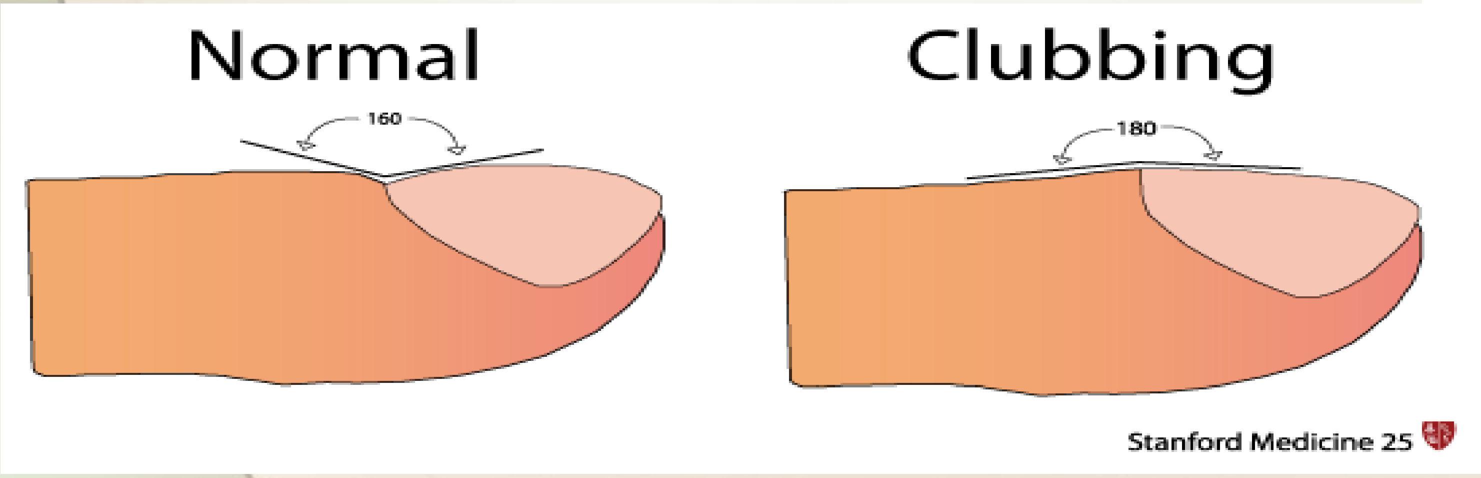
- Heart and Lung diseases
- Gastrointestinal and Liver diseases (malabsorption, crohns, colitis, cirrhosis)
- Graves Diseases
- Malignancy

*red line = clubbing

clubbing

- Angle between nail plate and fold greater than 180 degrees





Onychomatricoma (tumour of the nail matrix)

- Make sure to biopsy this one as it can mimic many other pathologies

Acral lentiginous melanoma

- Seen on palms, soles, mouth but when occurs within the nail a clue is that it involves the periungal region

Miscellaneous

Periungal

Periungal telangiectasias: strong association with CVD: SLE, dermatomyositis, scleroderma

Mucus cyst

Fibroma

Herpetic Whitlow